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M1: Applied Mathematics & Statistics for Computational Sciences

September 11, 2025

- 1 Overview: AI and ML
- 2 Refresher: Algebraic Foundations
- 3 Vectors and Vector Operations
- 4 Vector Spaces
- 5 From Math to ML Pipelines
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AI (Artificial Intelligence): systems that perform tasks usually requiring human intelligence:

- **Language understanding:** e.g., *Google Translate* converting French → English.
- **Decision-making:** e.g., *autonomous cars* detecting pedestrians and deciding when to stop.

ML (Machine Learning): data-driven methods within AI that learn patterns from data:

- **Email spam filters:** trained on labeled messages (“spam” vs “not spam”).
- **Recommendation systems:** e.g., *Netflix* predicting shows you may like.

AI is the broader ambition; ML is the statistical and algorithmic toolkit that makes it work.

Linear algebra:

- Images are matrices of pixel values.
- Neural nets apply linear transforms $Wx + b$ at each layer.
- PCA reduces dimensionality (e.g., compressing face images).

Calculus:

- Training $\approx$ minimizing a loss function.
- Gradient descent updates weights via derivatives.
- Example: adjusting millions of parameters in deep networks.

Probability & Statistics:

- Outputs are probabilities (e.g., disease detection with 85% confidence).
- Hypothesis testing $\rightarrow$ decide if improvements are significant.
- Bayesian models quantify uncertainty in predictions.

Face recognition: Image $\rightarrow$ vector (linear algebra) $\Rightarrow$ train weights with gradients (calculus) $\Rightarrow$ output a similarity score with confidence (probability/statistics).

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Definition 1

A function $f : X \rightarrow Y$ assigns each $x \in X$ a unique $y = f(x) \in Y$.

Examples

- Linear: $f(x) = ax + b$ (e.g., linear regression).
- Quadratic: $f(x) = ax^2 + bx + c$ (optimization/control).
- Exponential: $f(x) = ae^{bx}$ (growth/decay, LR schedules).
- Linear: $3x + 2 = 0 \Rightarrow x = -\frac{2}{3}$
- Quadratic: $ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

In ML: solving equations $\approx$ estimating parameters that satisfy optimality conditions (normal equations, gradients = 0).

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Definition 2

A vector $\mathbf{v} = [v_1, \dots, v_n] \in \mathbb{R}^n$.

Operations

- Addition: $\mathbf{a} + \mathbf{b} = [a_1 + b_1, \dots, a_n + b_n]$
- Scalar mult.: $c\mathbf{v} = [cv_1, \dots, cv_n]$
- Dot product: $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$

Recommendation systems: user/movie vectors $\Rightarrow$ rating $\approx \mathbf{u} \cdot \mathbf{m}$.

- Angle: $\cos \theta = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\|\mathbf{a}\| \|\mathbf{b}\|}$
- Orthogonality: $\langle \mathbf{a}, \mathbf{b} \rangle = 0$
- Cauchy–Schwarz: $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$

Cosine similarity is normalized inner product — ubiquitous in IR/NLP.

Given $\mathbf{a} = (1, 2, 2)$ and $\mathbf{b} = (2, 0, 1)$:

- 1 Compute $\mathbf{a} \cdot \mathbf{b}$ and the angle.
- 2 Are they orthogonal?
- 3 Normalize both and recompute the dot product (cosine similarity). Do it with Python.

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Definition 3

A **vector space** V over a field F has operations $u + v \in V$ and $a \cdot v \in V$ ($a \in F$) satisfying the usual axioms (associativity, commutativity of addition, distributivity, identities, inverses).

Example: $\mathbb{R}^3$ is a vector space over $\mathbb{R}$.

- **Linear independence:** $\sum_i \alpha_i \mathbf{v}_i = \mathbf{0} \Rightarrow \alpha_i = 0 \forall i$.
- **Span:** all linear combinations of a set.
- **Basis:** independent set that spans the space.
- **Dimension:** # vectors in a basis.

In $\mathbb{R}^2$, $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ form a basis $\Rightarrow \dim \mathbb{R}^2 = 2$.
 PCA finds a low-dimensional basis capturing most variance.

- Stack vectors as columns in A and compute $\text{rank}(A)$.
- If $\text{rank}(A) = k$ (for k vectors), they are independent.
- Numerically: use SVD; beware near-dependencies (conditioning).

Consider

$$\mathbf{v}_1 = (1, 0, 1), \quad \mathbf{v}_2 = (0, 1, 1), \quad \mathbf{v}_3 = (1, 1, 2).$$

- 1 Are they independent? (hint: rank)
- 2 Find a basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.
- 3 What is the dimension of that span? Do it with Python.

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- **Representation:** features $\rightarrow$ vectors/matrices.
- **Objective:** define loss (often quadratic, convex locally).
- **Optimization:** gradient-based updates.
- **Evaluation:** statistical metrics and uncertainty.

Strong math foundations $\Rightarrow$ interpretability, robustness, and reproducibility.

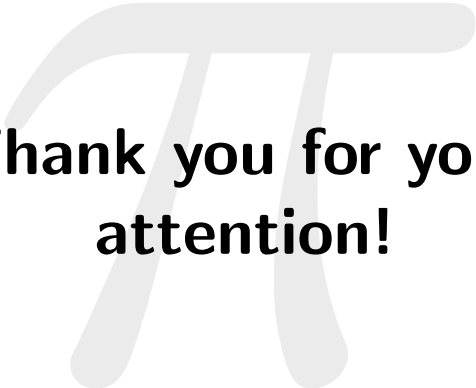
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- 1 Refreshed algebra (functions, solving equations).
- 2 Built core objects: vectors, vector spaces, independence/basis/dimension.
- 3 Connected math to practice (representation, optimization, evaluation).

Next: hands-on practice in Python (NumPy) to operationalize these ideas.

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**Thank you for your
attention!**